CANARA COLLEGE LIBRARY MANGALURU-3

Reg. No. | | | | | | | | | | | |

BCAAIMS 103

First Semester B.C.A. (AI&ML) Degree Examination, December 2025/January 2026 (SEP) (2024-25 Batch Onwards) COMPUTATIONAL MATHEMATICS

Time : 3 Hours Max. Marks : 80

Note : *Answer* ***any ten*** *questions from Part –* ***A*** *and* ***any one full*** *question from* ***each*** *Unit of Part –* ***B.***

PART – A

(2×10=20)

1. a) Find the value of $\log_4 256$.

   b) Prove that $\log_y x.\log_z y.\log_x z = 1$.

   c) Find the co-ordinates of the centroid of the triangle whose vertices are (2,3), (–5, 2) and (1,7).

   d) Find the centre and radius of the circle whose equation is $x^2 + y^2 - 4x + 8y - 5 = 0$.

   e) Define skew-symmetric matrix with an example.

   f) If matrix $A = \begin{bmatrix} 4 & -3 \\ 8 & -2 \\ -1 & 0 \end{bmatrix}$ find 4A.

   g) Construct the truth tables for $P \wedge (P \vee Q)$.

   h) Define disjoint sets. Give example.

   i) Write the power set of A = {1, 2, 3}.

   j) Let x = {1, 5, p, Jack} ,Y = {2, 5, 7, q, Jill} and f = {<1, 2>,<5,7>,<p, q>,<Jack, q>}, Find Df and Rf.

   k) What is Pigeonhole principle ? Give an example.

   l) Define strongly and weakly connected graph.

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PART B

Unit – I

2. a) Prove that $\frac{\log\sqrt{27}+\log\sqrt{8}-\log\sqrt{125}}{\log 6-\log 5}=\frac{3}{2}$.

   b) Expand $(3x - y)^4$ using Binomial theorem.

   c) Find the coordinates of a point which divides the line joining two points P(8, 9) and Q(–7, 4) internally in the ratio 2:3 and externally in the ratio 4:3. **(5+5+5)**

3. a) Show that the points (–1, 2), (3, –1) and, (2, 6) form an isosceles triangle.

   b) Find the equation of straight line parallel to $2x - 3x - 5 = 0$ and passing through (4, 5).

   c) Find the equation of tangent to the circle $x^2 + y^2 - 6x + 8y + 17 = 0$ at (1, –2). **(5+5+5)**

Unit – II

4. a) If $A=\begin{bmatrix}1 & 2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix}$ show that $A^3 - 3A^2 - A + 9I = 0$.

   b) Solve the following simultaneous linear equations using Cramer's Rule.

   $2x + y - z = 3$

   $x + y + z = 1$

   $x - 2y - 3z = 4$

   c) A man borrows Rs. 5,115 to be paid in 10 monthly instalments. If each instalment is double the value of the last, find the value of the first and last instalments. **(5+5+5)**

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5. a) Compute the Inverse of a matrix $\begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$.

b) A firm produces 1000 sets of TV during its first year. The total sets produced at the end of 5 years are 7000. Estimate the annual rate of increase in production, if the increase in each year is uniform.

c) Find three numbers in GP whose product is 1000 and sum is 35. **(6+4+5)**

Unit – III

6. a) Show the following implications :

$(P \wedge Q) \Rightarrow (P \rightarrow Q)$

$P \Rightarrow (Q \rightarrow P)$

b) $A = \{1, 2, 3\}$, $B = \{1, 2, 5, 7, 9\}$ write $A - B$, $B - A$, $A + B$, $A \cup B$, $A \cap B$.

c) $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (4, 4)\}$ construct relation matrix of R and draw digraph of R. **(5+5+5)**

7. a) $A = \{1\}$, $B = \{a, b\}$, $C = \{2, 3\}$ write $A \times B$, A^2, $A^2 \times B$, $A \times B \times C$.

b) Let $x = \{1, 2, 3\}$, f, g, h and s are the functions from X to X given by

$f = \{(1, 2), (2, 3), (3, 1)\}$ $h = \{(1, 1), (2, 2), (3, 1)\}$

$g = \{(1, 2), (2, 1), (3, 3)\}$ $s = \{(1, 1), (2, 2), (3, 3)\}$

Find $f \circ g$, $g \circ f$, $s \circ s$, $s \circ g$ and $f \circ s$.

c) Given the relation matrices $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ $Ms = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$

Find $M_{R \circ S}$, $M_{\tilde{R}}$, $M_{\tilde{S}}$, $M_{\widetilde{R \circ S}}$ and show that $M_{\widetilde{R \circ S}} = M_{\tilde{S}} \circ M_{\tilde{R}}$. **(5+5+5)**

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Unit – IV

8. a) There are 18 Mathematics majors and 325 computer science majors at a college. In how many ways can two representatives be picked so that one is a mathematics major and the other is a computer science major.